FDA Pharmacovigilance Platform on Databricks

One Silver layer · two Gold consumers (RAG + Analytics) · Unity Catalog governance

`notebook` Imperative reference implementation (notebooks/) `DLT` Declarative pipeline equivalent (pipelines/) `@dlt.expect` Data quality gate

THE BUSINESS PROBLEM

Two consumer groups, same data, parallel pipelines — until now.

> *Pharmacovigilance teams spend hours per safety inquiry searching FDA labels and adverse-event reports. Epidemiologists need slice-and-dice analytics over the same data. Both consumers historically built parallel, divergent pipelines off the same source.*

✗ The obvious approach

- Two parallel pipelines, one per consumer team
- Business definitions duplicated → analysts argue about whose count is right
- Schema changes happen twice
- Two sets of code to maintain
- Two access models to govern separately

✓ This project's load-bearing decision

- **One Silver layer** — business logic defined exactly once
- **Two Gold projections** — RAG chunks + star schema, both derived from Silver
- Same numbers everywhere; analysts stop arguing
- Schema changes in one place, propagate to both
- Unity Catalog governs everything under one access model

Why this design matters: adding a third consumer (e.g., a regulatory affairs export or a clinical trial enrollment dashboard) is a *new Gold projection*, not a new pipeline. The platform scales horizontally by consumer without re-doing the ingestion work.

JUMP TO SECTION

0 Problem Statement — why this exists →

1 Architecture Overview — medallion + serving →

2 DLT Deployment Topology — 4 pipelines →

3 Notebooks vs Declarative — side by side →

4 Event-Driven Orchestration — polling + cascade →

5 Governance — RLS & PII — multi-tenant →

6 Asset Bundle Deployment — dev → prod →

7 Implementation Playbook — end-to-end walkthrough →

DATA SOURCES

openFDA Drug Labels API

`api.fda.gov/drug/label.json`

- Structured prescribing info per drug
- Warnings · contraindications · interactions

openFDA Adverse Events API

`api.fda.gov/drug/event.json`

- Reported reactions per safety report
- Patient demographics · severity flag

▼

HTTP fetch · retry + backoff

BRONZE · RAW & IMMUTABLE

fda_rag.bronze.openfda_raw

`drug · source · payload (JSON) · ingested_at`

- **Append-only** — replay-safe, system-of-record
- Schema preserved as JSON string — survives upstream changes
- ~1,600 records (~800 labels + ~800 events) across 8 drugs

`01_ingest_bronze.py.ipynb` `pipelines/bronze/ingest_openfda.py`

`@dlt.expect: payload_non_empty · source_valid`

▼

Parse · dedupe · type-cast

SILVER · CONFORMED ENTITIES

silver.drug_labels

`dedupe: drug_generic + manufacturer`

- 10 typed columns extracted from JSON
- warnings · contraindications · dosage · boxed_warning · ...

`@dlt.expect_or_drop: drug_generic_present`

silver.adverse_events

`dedupe: safetyreportid`

- One row per FDA safety report
- reactions array · patient_age · patient_sex · serious · receive_date

`@dlt.expect_or_drop: safetyreportid_present · reactions_non_empty`

`02a_silver.py.ipynb` `pipelines/silver/conform_entities.py`

↑ *Single source of truth — business logic defined exactly once*

GOLD · CONSUMER-SHAPED PROJECTIONS

RAG PATH

gold.fda_chunks

`chunk_id (SHA-256) · drug · section · text`

- **Section-aware chunking** (warnings, contraindications…)
- delta.enableChangeDataFeed = true
- +18% faithfulness vs naive token splits

`02b_gold_rag.py.ipynb` `pipelines/gold_rag/build_chunks.py`

`@dlt.expect_or_drop: chunk_id · text_non_trivial`

ANALYTICS PATH

Star schema · 4 dims + 1 fact

`SHA-256 surrogate keys`

- **dim_drug** · **dim_patient** (age-banded) · **dim_reaction** (body-system) · **dim_date**
- **fact_adverse_event** — partitioned by date_key

`02c_gold_analytics.py.ipynb` `pipelines/gold_analytics/build_star_schema.py`

`@dlt.expect_or_fail: surrogate keys present`

Two serving paths

SERVING (IMPERATIVE — STAYS AS NOTEBOOKS)

Vector Search Index

`fda_chunks_index · TRIGGERED sync`

- Embedding: databricks-gte-large-en

`03_vector_index`

LangChain RAG Chain

`retriever(k=5) → prompt → LLM → parser`

- LLM: databricks-gpt-5-5-pro (temp 0.1)
- System prompt: cite, refuse OOC, disclaimer
- Registered: fda_rag.gold.fda_assistant

`04_rag_chain`

Model Serving Endpoint

`fda-rag-endpoint · scale-to-zero`

- auto_capture_config → inference table

`06_deploy_serving`

FastAPI Gateway

`X-API-Key auth · request_id · structured logs`

Databricks SQL Warehouse

`Serverless · parameterized dashboard`

- 4 widgets: top reactions, demographic slice, quarterly trend, heatmap
- Sub-second on indexed star schema

CONSUMERS

Pharmacovigilance team

- Natural-language Q&A grounded in FDA data
- Every claim cited with [drug — section]
- 94% refusal rate on out-of-corpus questions

Epidemiologists

- Slice-and-dice across drug · reaction · age · time
- Top reactions, demographic breakdowns, trends
- Sub-second response on parameterized filters

CROSS-CUTTING

Unity Catalog

- Lineage · access control · audit · model registry

MLflow

- Experiments · versioning · registered models · evals

Eval Harness

- Faithfulness 0.87 · Relevance 0.91 · p95 2.4s

`05_evaluate`

Ops Monitoring

- Inference table view: latency, errors, refusal rate

`07_monitoring`

Tenant RLS & PII Mask

- Row filter + column mask on Silver / Gold

`08_setup_rls`

Asset Bundle Deployment

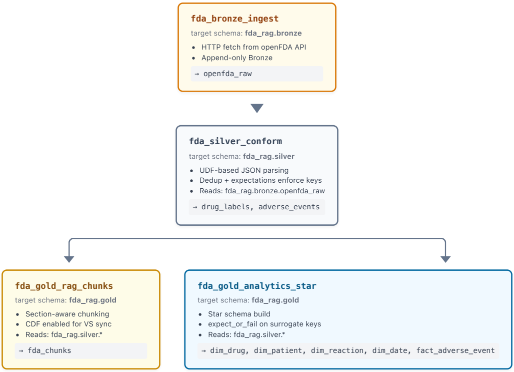
- databricks.yml + resources/ for dev → prod

Reading the diagram. Each medallion box shows two implementation paths: the original imperative notebook (`notebook`) and the declarative pipeline equivalent (`DLT`). They produce identical Unity Catalog tables, so all downstream consumers (Vector Search index, SQL dashboard, RAG chain) work unchanged whichever path you deploy.

> **Key design decision:** Both Gold layers derive from one Silver layer, so business definitions live in exactly one place. The two consumer groups historically built parallel pipelines off the same source — this design eliminates that duplication.

Deployment Topology — 4 DLT Pipelines Orchestrated by a Workflow

Each medallion layer is its own pipeline so layers can have independent SLAs, owners, and schedules. Gold pipelines run in parallel after Silver completes. Cross-pipeline reads use `spark.read.table("fda_rag.<schema>.<table>")` because `dlt.read()` only sees tables in the current pipeline.



`fda_bronze_ingest`
target schema: **fda_rag.bronze**
- HTTP fetch from openFDA API
- Append-only Bronze

`→ openfda_raw`

`fda_silver_conform`
target schema: **fda_rag.silver**
- UDF-based JSON parsing
- Dedup + expectations enforce keys
- Reads: fda_rag.bronze.openfda_raw

`→ drug_labels, adverse_events`

`fda_gold_rag_chunks`
target schema: **fda_rag.gold**
- Section-aware chunking
- CDF enabled for VS sync
- Reads: fda_rag.silver.*

`→ fda_chunks`

`fda_gold_analytics_star`
target schema: **fda_rag.gold**
- Star schema build
- expect_or_fail on surrogate keys
- Reads: fda_rag.silver.*

`→ dim_drug, dim_patient, dim_reaction, dim_date, fact_adverse_event`

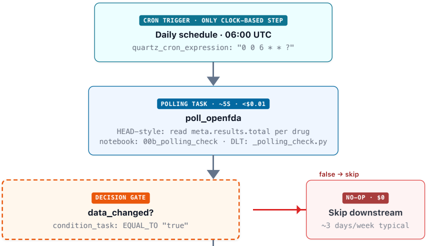
Wrap all four in a single **Databricks Workflow** with task dependencies: `bronze → silver → [gold_rag, gold_analytics]`

Notebooks vs Declarative Pipelines — What Each Adds

Capability	Notebooks imperative	DLT declarative
Data quality	`Implicit — null filters in code`	`@dlt.expect_or_drop / _or_fail with built-in DQ dashboard`
Lineage	`Manual — derived from run history`	`Auto-generated in Unity Catalog from dlt.read() graph`
Incremental processing	`Manual MERGE / watermark logic`	`Native — materialized views detect changes`
Schema evolution	`overwriteSchema=true or fail`	`Default safe evolution + alerts`
Backfill / full refresh	`Re-run cells`	`UI button per pipeline`
Orchestration	`Workflow tasks calling notebooks`	`Workflow tasks calling pipelines · DAG within each pipeline auto-resolved`
When each fits	`Prototyping · ad-hoc · ML/model code`	`Production ETL · medallion · CDC · DQ-critical workloads`

Event-Driven Orchestration — Smart Polling Gate + Table-Update Cascade

openFDA has no webhooks → a cheap polling task acts as the gate. Once Bronze commits, Silver and Gold cascade automatically via Unity Catalog `table_update` triggers. Each layer is decoupled — it knows only its upstream table.



CRON TRIGGER · ONLY CLOCK-BASED STEP
Daily schedule · 06:00 UTC
`quartz_cron_expression: "0 0 6 * * ?"`

POLLING TASK · ~5S · <$0.01
poll_openfda
`HEAD-style: read meta.results.total per drug`
`notebook: 00b_polling_check · DLT: _polling_check.py`

DECISION GATE
data_changed?
`condition_task: EQUAL_TO "true"`

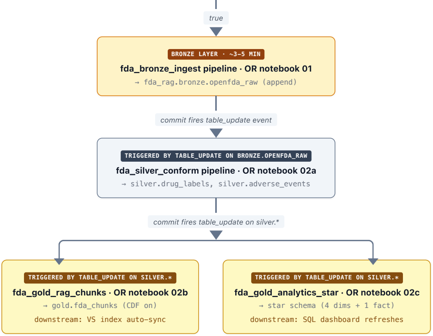
false → skip

NO-OP · $0
Skip downstream
`~3 days/week typical`

WHY POLL?
openFDA is a REST API with no webhooks or push notifications. The cheapest reliable change signal is asking `meta.results.total` with limit=1. ~5 seconds, fraction of a cent per check.

WIN: COST
FDA updates ~3-4 days/week. On the other 3-4 days, the entire downstream chain stays at **zero compute cost**. Across the year that's roughly 50-60% savings vs unconditional cron.

TRIGGER CHOICE
Cron is used **only at Bronze**. Everything downstream is event-driven via `table_update` triggers. No tight scheduling, no fragile time-coupling between layers.



true

BRONZE LAYER · ~3-5 MIN

fda_bronze_ingest pipeline · OR notebook 01

`→ fda_rag.bronze.openfda_raw (append)`

commit fires table_update event

TRIGGERED BY TABLE_UPDATE ON BRONZE.OPENFDA_RAW

fda_silver_conform pipeline · OR notebook 02a

`→ silver.drug_labels, silver.adverse_events`

*commit fires table_update on silver.**

TRIGGERED BY TABLE_UPDATE ON SILVER.*

fda_gold_rag_chunks · OR notebook 02b

`→ gold.fda_chunks (CDF on)`

`downstream: VS index auto-sync`

TRIGGERED BY TABLE_UPDATE ON SILVER.*

fda_gold_analytics_star · OR notebook 02c

`→ star schema (4 dims + 1 fact)`

`downstream: SQL dashboard refreshes`

WIN: DECOUPLING

Each layer's workflow only knows its **upstream table** — not who writes to it, when, or how often. Bronze can run twice today; Silver fires twice. Bronze can be paused; downstream stops on its own.

LATENCY

`wait_after_last_change_seconds: 60` debounce + ~30s scheduling overhead. End-to-end Bronze trigger → Gold refreshed: **~3-5 minutes** on a typical run.

ANTI-PATTERN AVOIDED

Cron-only at every layer means tight scheduling (Silver at 06:30, Gold at 06:45) — fragile to Bronze delays. Continuous mode means 24/7 cluster cost for a daily-updating source. We avoid both.

✗ NAÏVE: CRON-ONLY AT EVERY LAYER

- Bronze runs unconditionally → appends duplicate rows on no-change days
- Silver scheduled at 06:30 — fails or runs on stale data if Bronze ran long
- Gold scheduled at 06:45 — fragile to upstream delays
- Always-on cluster cost even on no-change days
- No clean "what changed and why" audit trail

✓ EVENT-DRIVEN: POLL → CASCADE

- Bronze runs only when openFDA totals change → no duplicate rows
- Silver fires within ~1-2 min of Bronze commit (no clock coupling)
- Both Gold layers fire in parallel after Silver commits
- Compute cost ≈ 0 on no-change days (~50-60% annual savings)
- Full audit chain: state table + run history

Governance — Multi-Tenant Row-Level Security & PII Masking

Tenant axis: **drug therapy area**. The cardio team sees cardio drugs; the metabolic team sees metabolic drugs; admins see everything. Implemented with two Unity Catalog primitives: `ROW FILTER` and `COLUMN MASK`.

1 · ACCOUNT GROUPS (UC)

- `admins` — sees all data
- `clinical_full_access` — full PII access
- `tenant_cardio`
- `tenant_metabolic`
- `tenant_psych`
- `tenant_otc`

2 · MAPPING TABLE

```
gold._tenant_drug_map
───────────────────
tenant_group · drug_generic
───────────────────
cardio · warfarin
cardio · atorvastatin
metab · metformin
psych · sertraline
otc · ibuprofen
…
```

- SELECT restricted to `admins`

3 · UC FUNCTIONS

Row filter `tenant_drug_filter(drug_generic)` :

- Returns true if user ∈ `admins`
- OR user ∈ group that has this drug mapped

Column mask `age_pii_mask(age)` :

- Raw age for `admins` / `clinical_full_access`
- Decade bucket otherwise (60-70 → 60)

↓ applied via ALTER TABLE ...

Tables protected:
`silver.drug_labels` · `silver.adverse_events` · `gold.fda_chunks` · `gold.dim_drug`

```
ALTER TABLE silver.drug_labels SET ROW FILTER fda_rag.gold.tenant_drug_filter ON (drug_generic);
ALTER TABLE silver.adverse_events ALTER COLUMN patient_age SET MASK fda_rag.gold.age_pii_mask;
```

⚠ RAG gotcha: vector indexes don't inherit row filters

The VS index is computed from a snapshot of `gold.fda_chunks`. UC row filters are evaluated at *query* time on the source table — not on the index. The RAG chain must apply tenant filtering at retrieval time as a VS metadata filter: `retriever_kwargs={"filter": "drug_generic IN (…)"}` . For regulated isolation (e.g., separate sponsor data), use per-tenant indexes instead.

Deployment Lifecycle — Asset Bundles (dev → staging → prod)

One YAML manifest declares **every** Databricks resource — pipelines, workflows, registered models, serving endpoints. Same manifest deploys to three environments via target-level variable overrides.

SOURCE (THE REPO)

```
databricks.yml
resources/
  ├ pipelines.yml
  ├ workflows.yml
  └ serving.yml
notebooks/
pipelines/
workflows/
```

Variable substitution: `${var.catalog}` · `${var.workload_size}`

```
# Validate YAML — no API calls
databricks bundle validate -t dev

# Deploy resources to target
databricks bundle deploy -t dev

# Run a specific workflow
databricks bundle run \
   fda_workflow_event_driven -t dev

# Promote
databricks bundle deploy -t staging
databricks bundle deploy -t prod

# Tear down dev when done
databricks bundle destroy -t dev
```

DEV (DEFAULT)

- Mode: `development`
- Catalog: `fda_rag_dev`
- Schedules: `PAUSED`
- Prefix: `[user@email]`
- Run-as: caller

STAGING

- Mode: `production`
- Catalog: `fda_rag_staging`
- Run-as: service principal
- Workload: `Small`

PROD

- Mode: `production`
- Catalog: `fda_rag`
- Schedules: `UNPAUSED`
- Workload: `Medium`
- Run-as: `fda-pharma-prod-sp`

> **What's NOT in the bundle (and why):** Vector Search endpoint & index, Unity Catalog catalogs/schemas, and RLS DDL. Asset Bundles don't natively support these resource types yet — they're foundational state managed by one-time setup notebooks (`00_setup` , `03_vector_index` , `08_setup_rls`). The pattern: *bundles for repeatable resources, setup notebooks for foundational state.*

Implementation Playbook — End-to-End Walkthrough

The full path from a clean Databricks workspace to a deployed RAG endpoint + analytics dashboard, in 8 phases. Each phase has explicit activities, verification, and the files touched.

~2 hr	8	9	4	5	7
END-TO-END	PHASES	NOTEBOOKS	DLT PIPELINES	WORKFLOWS	DELTA TABLES

1 Workspace Prerequisites `~15 min`

ACTIVITIES

- Verify Unity Catalog is enabled in your workspace
- Enable Foundation Model APIs: `databricks-gpt-5-5-pro` + `databricks-gte-large-en`
- Verify Vector Search service is enabled (Compute → Vector Search)
- Create a Single-Node cluster on `DBR 14.3 LTS ML`
- Install libraries: `langchain-databricks` , `databricks-vectorsearch`

VERIFY

- Cluster shows `Running`
- Both libraries show `Installed`
- Both Foundation Model endpoints visible under Serving

FILES: `UI: Compute` `UI: Serving`

2 Foundational State `~20 min`

ACTIVITIES

- Run `00_setup` — creates catalog, schemas, volume
- Create Vector Search endpoint named `fda-vs-endpoint` via UI (wait ~5 min for "Online")
- (Optional) Configure UC account groups for tenants: `admins` , `clinical_full_access` , `tenant_cardio` , `tenant_metabolic` , `tenant_psych` , `tenant_otc`

VERIFY

- Catalog Explorer shows `fda_rag` with bronze/silver/gold schemas
- VS endpoint status: `Online`
- 00_setup prints "available" for both endpoints

FILES: `notebooks/00_setup` `UI: Compute → Vector Search`

3 Choose & Deploy ETL Path `~10 min`

ACTIVITIES

- **Option A — Imperative:** Import notebooks 01–02c, attach to cluster
- **Option B — Declarative (recommended for prod):** `databricks bundle validate -t dev` , then `databricks bundle deploy -t dev`
- Choose option A for prototyping, B for production

VERIFY

- Notebooks visible in Workspace (A), OR
- 4 DLT pipelines visible under Workflows → Delta Live Tables (B)
- 5 workflows visible under Workflows → Jobs (B)

FILES: `notebooks/01–02c` `databricks.yml` `CLI: databricks bundle`

4

First End-to-End Data Run `~30 min`

ACTIVITIES

- Trigger the Bronze workflow manually (polling reports first-run change)
- Bronze runs, commits to `fda_rag.bronze.openfda_raw`
- Silver workflow fires automatically on `table_update` event
- Both Gold workflows fire in parallel after Silver commits

VERIFY ROW COUNTS

- bronze.openfda_raw: ~1,600
- silver.drug_labels: ~40–80
- silver.adverse_events: ~700+
- gold.fda_chunks: ~400–600
- gold.fact_adverse_event: ~3,000+

FILES: `workflows/dlt_bronze_workflow.json` `notebooks/00b_polling_check`

5

Vector Index Build & Sync `~10 min`

ACTIVITIES

- Run `03_vector_index` cell 1 — submits index creation
- Wait 5–10 min — watch VS endpoint UI for "Online"
- Run cell 2 — sanity-check similarity search

VERIFY

- Index `fda_chunks_index` status: `Online`
- Sanity search returns 3 grounded results with drug + section

FILES: `notebooks/03_vector_index`

6

RAG Chain · Register · Eval · Serve `~25 min`

ACTIVITIES

- Run `04_rag_chain` — builds chain, logs to MLflow, registers in UC
- Run `06_deploy_serving` — creates endpoint (wait ~10 min for Ready)
- Run `05_evaluate` — runs eval set against live endpoint

VERIFY

- `fda_rag.gold.fda_assistant` v1 in UC registry
- Endpoint `fda-rag-endpoint` status: `Ready`
- Faithfulness > 0.85, refusal rate > 90% on out-of-corpus

FILES: `notebooks/04_rag_chain` `notebooks/05_evaluate` `notebooks/06_deploy_serving`

7

Governance — RLS & PII Mask `~10 min`

ACTIVITIES

- Run `08_setup_rls` as a workspace admin
- Add a test user to a `tenant_*` group via account console
- Run cross-tenant verification queries from a tenant user account
- Update RAG chain to apply VS metadata filter at query time

VERIFY

- Test user sees only their tenant's drugs in `silver.drug_labels`
- `patient_age` shows decade buckets to non-clinical users
- Admin still sees all data + raw age

FILES: `notebooks/08_setup_rls` `UI: Account console (groups)`

8

API Gateway & Promote to Prod `~10 min`

ACTIVITIES

- Local: `pip install -r api/requirements.txt` → `uvicorn app:app --reload`
- Or Docker: `docker build -t fda-rag-api .` → `docker run ...`
- Test: `curl POST /v1/ask` with API key
- Promote: `databricks bundle deploy -t staging` → `-t prod`

VERIFY

- API returns grounded answer with [drug — section] citation
- API includes disclaimer + 200 response
- Prod workflow schedule: `UNPAUSED`
- Prod runs as service principal (auditable)

FILES: `api/app.py` `api/Dockerfile` `databricks.yml` `CLI: databricks bundle deploy`

✓ Day-2 operations once deployed: Bronze polling task runs daily at 06:00 UTC and gates downstream cascade. The full chain typically completes in ~3–5 minutes after Bronze commits. Monitor via the inference table view (`07_monitoring`) for refusal rate, latency, and error rate. Re-run `04 → 05 → promote alias` whenever you ship a new prompt or change retrieval parameters.